



SIMATS ENGINEERING

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Course Code:CSA1522

Slot:C

Course Name:CLOUD COMPUTING AND BIG ANALYTICS

Course Faculty:Dr.P.RAJARA,

Project Title:HDFS-BASED LARGE -SCALE HEALTHCARE DATA STORAGE AND RETRIEVAL SYSTEM

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description:

Module 1: Distributed Data Storage (HDFS)

The Hadoop Distributed File System (HDFS) is a distributed storage system designed to store and manage very large amounts of data across multiple computers in a Hadoop cluster. It splits large files into smaller data blocks and distributes them across DataNodes, while the NameNode manages file locations and metadata. HDFS creates multiple copies of each data block to ensure fault tolerance and prevent data loss if a node fails. This architecture provides high scalability, reliability, and efficient data access, making HDFS an ideal solution for big data applications such as healthcare, banking, and social media analytics. A NameNode manages file metadata, while DataNodes store the actual data, making HDFS suitable for handling big data efficiently.

Student Signature

Guide Signature